

AGGREGATE PLANNING

AGGREGATE PLANNING-1

- ◆ Organizations cannot predict exactly the quantity and timing of demands for specific products or services months in advance under these conditions.
- ◆ They typically must assess their capacity needs (e.g., labor, inventories) and costs months in advance in order to be able to handle demand.

Aggregate Planning

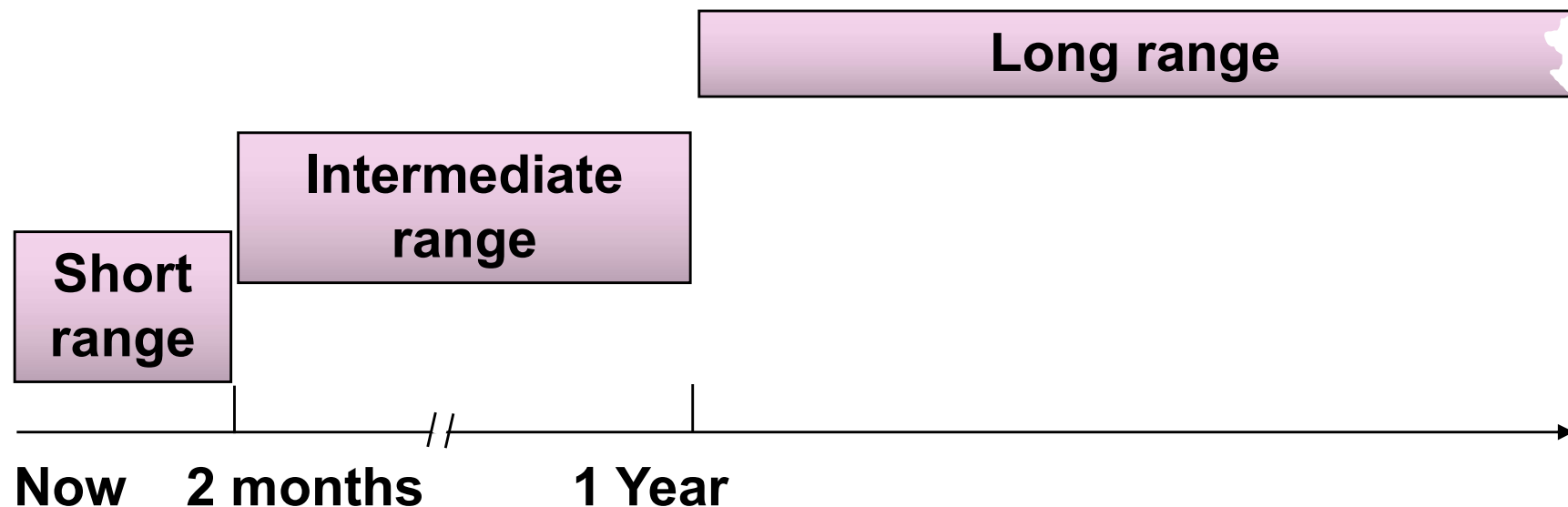
The objective of aggregate planning is to meet the forecasted demand while minimizing the total cost over the planning period

Aggregate planning

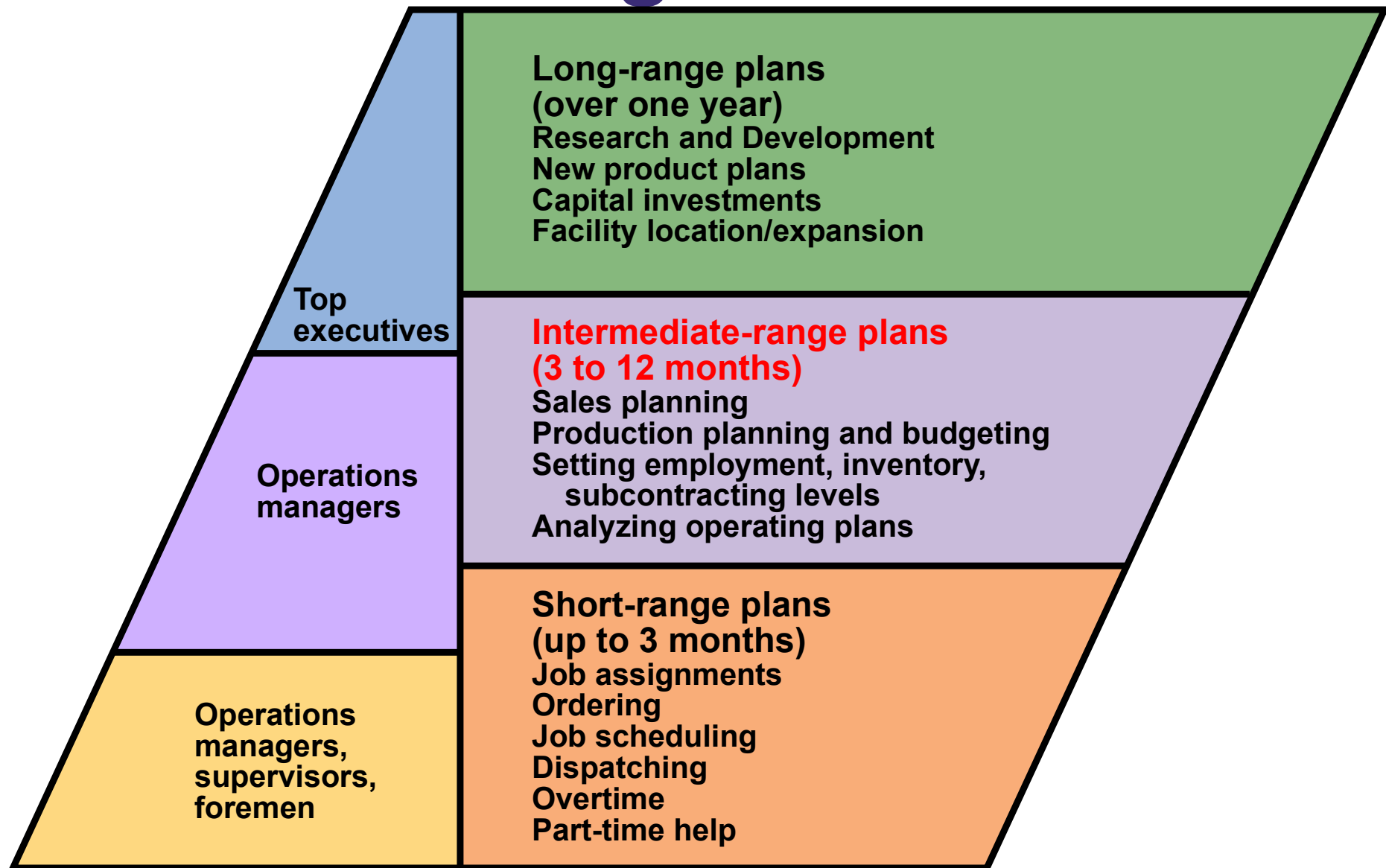
- ◆ Organizations make capacity decisions on three levels: long term, intermediate term, and short term.
- ◆ Long-term decisions relate to product and service selection facility size and location, equipment decisions, and layout of facilities. These long-term decisions essentially establish the capacity constraints within which intermediate planning must function.
- ◆ Intermediate decisions, as noted above, relate to general levels of employment, output, and inventories, which in turn establish boundaries within which short-range capacity decisions must be made.
- ◆ Thus, short-term decisions essentially consist of deciding the best way to achieve desired results within the constraints resulting from long-term and intermediate-term decisions. Short-term decisions involve scheduling jobs, workers and equipment

Planning Horizon

Aggregate planning: Intermediate-range capacity planning, usually covering 3 to 12 months. The goal of aggregate planning is to achieve a production plan that will effectively utilize the organization's resources to satisfy expected demand.



Planning Horizons



Responsibility

Planning tasks and horizon

Figure 13.1

Aggregate planning

- ◆ **Aggregate planning decisions are strategic decisions that define the framework within which operating decisions will be made.**
- **They are the starting point for scheduling and production control systems.**
- **They provide input for financial plans;**
- **They involve forecasting input and demand management, and**
- **They may require changes in employment levels.**

Aggregate planning

For Example

At the end of the aggregate production planning exercise , a garments manufacturer may arrive at the following plan

❖ Produce at the rate of 9000 meters of cloth everyday during the months of January to march

Increase it to 11000 meters during April to august and change the production rate 10,000 meters during September to December

❖ Carry 10% of monthly production as inventory during the first nine months of production

❖ Work on a one-shift basis throughout the year with 20% overtime during July to October

Aggregate Planning

- ◆ Aggregate planning is a **big picture** approach to **production planning**.
- ◆ It is a **production plan** to meet the **demand** throughout the **year**.
- ◆ It is **not** concerned with **individual products**, but with a single **aggregate product** representing all products.
- ◆ For example, in a TV manufacturing plant, the aggregate planning does **not** go into **all models and sizes**. It only deals with a **single representative aggregate TV**.
- ◆ **All models are lumped together** and represent a **single product**; hence the term **aggregate planning** is used.

Aggregate Planning Inputs

- ◆ **Demand forecasting for aggregate unit**
- ◆ **Resources**
 - ◆ **Workforce/production rate**
 - ◆ **Facilities and equipment**
- ◆ **Policies**
 - ◆ **Subcontracting**
 - ◆ **Overtime**
 - ◆ **Inventory levels**
 - ◆ **Back orders**
- ◆ **Costs**
 - ◆ **Inventory carrying**
 - ◆ **Back orders**
 - ◆ **Hiring/firing**
 - ◆ **Overtime**
 - ◆ **Inventory changes**
 - ◆ **subcontracting**

Aggregate Planning Outputs

- ◆ **Total cost of a plan**
- ◆ **Projected levels of:**
 - ◆ **Inventory**
 - ◆ **Output**
 - ◆ **Employment**
 - ◆ **Subcontracting**
 - ◆ **Backordering**

AGGREGATION

- ◆ Aggregation is important because it is not possible to predict with accuracy the timing and volume of demand for individual items

The ways to aggregate:

- ◆ Aggregate units of output per month
- ◆ Dollar value of total monthly output
- ◆ Measures that relate to capacity such as labor hours

Aggregate Planning

Quarter 1		
Jan	Feb	Mar
150,000	120,000	110,000



Quarter 2		
Apr	May	Jun
100,000	130,000	150,000



Quarter 3		
Jul	Aug	Sep
180,000	150,000	140,000



The Planning Process

Determine the quantity and timing of production for the intermediate future

- ◆ **Objective is to minimize cost over the planning period by adjusting**
 - ◆ **Production rates**
 - ◆ **Labor levels**
 - ◆ **Inventory levels**
 - ◆ **Overtime work**
 - ◆ **Subcontracting rates**

Aggregate Planning

Requirements for aggregate planning:

- ◆ A logical overall (aggregate) unit for measuring sales and output
- ◆ A **forecast of demand** for an intermediate planning period in these aggregate terms
- ◆ A method for determining the **costs**
- ◆ **An aggregate planning strategy** that combines forecasts and costs so that scheduling decisions can be made for the planning period

Aggregate Planning Strategies

1. Level Production: Maintaining a steady rate of production while meeting variations in demand by a combination of options

- ✓ Using **part-timers, overtime, or idle time** to absorb changes
- ✓ Using **subcontractors** and maintain a stable workforce
- ✓ **Backordering**, satisfying the demand of one period in one or more periods later.

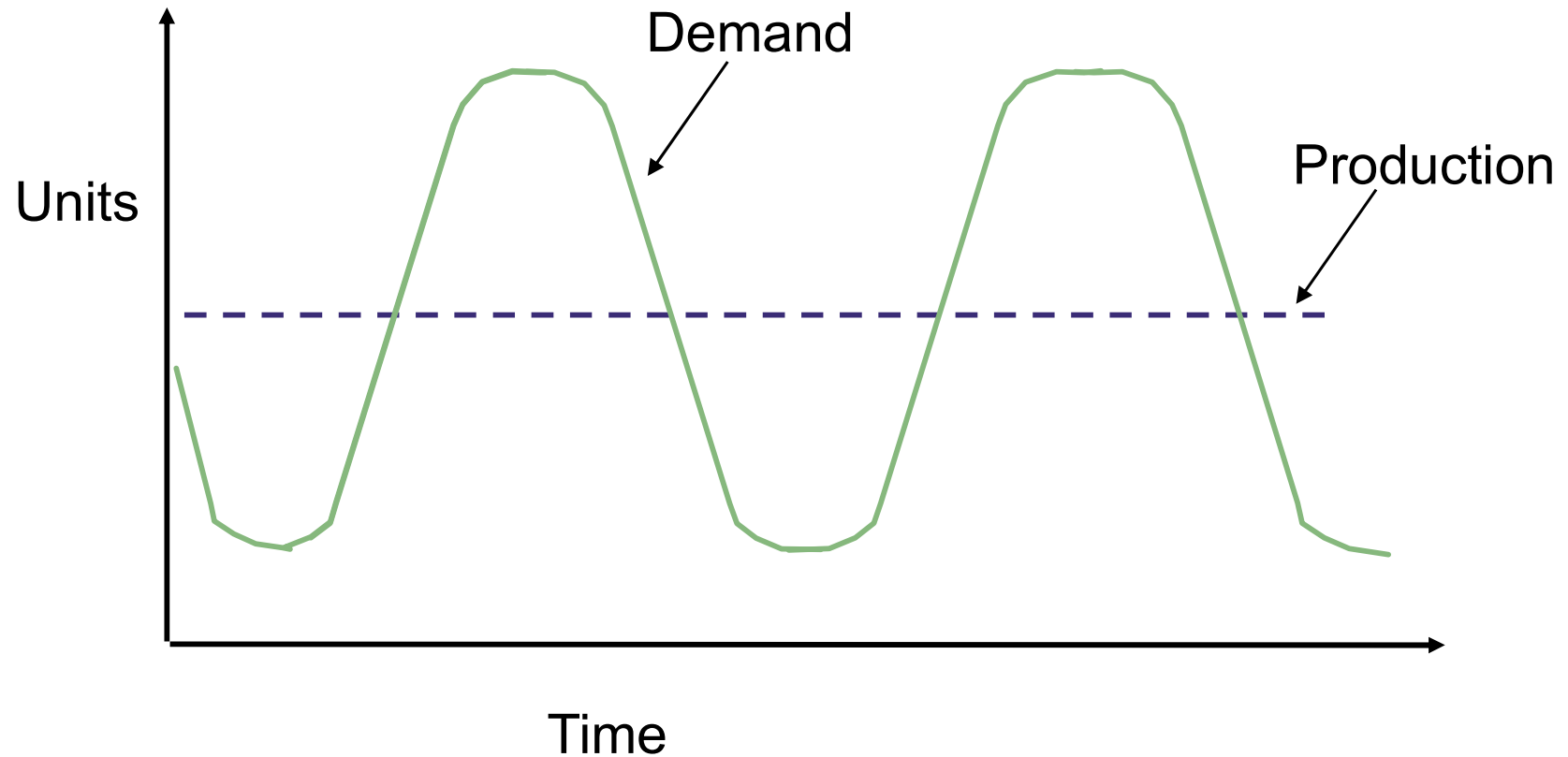
2. Chase Demand: Matching capacity to demand by keeping production in each period equal to the expected demand for that period.

3. Mixed Strategies

Level Production (Constant Workforce)

- ◆ Daily production is uniform
- ◆ Use inventory as buffer
- ◆ The underlying philosophy is that **stable employment leads to better quality, less turnover, less absenteeism, and more employee commitment.**
- ◆ **This strategy works well when demand is stable**

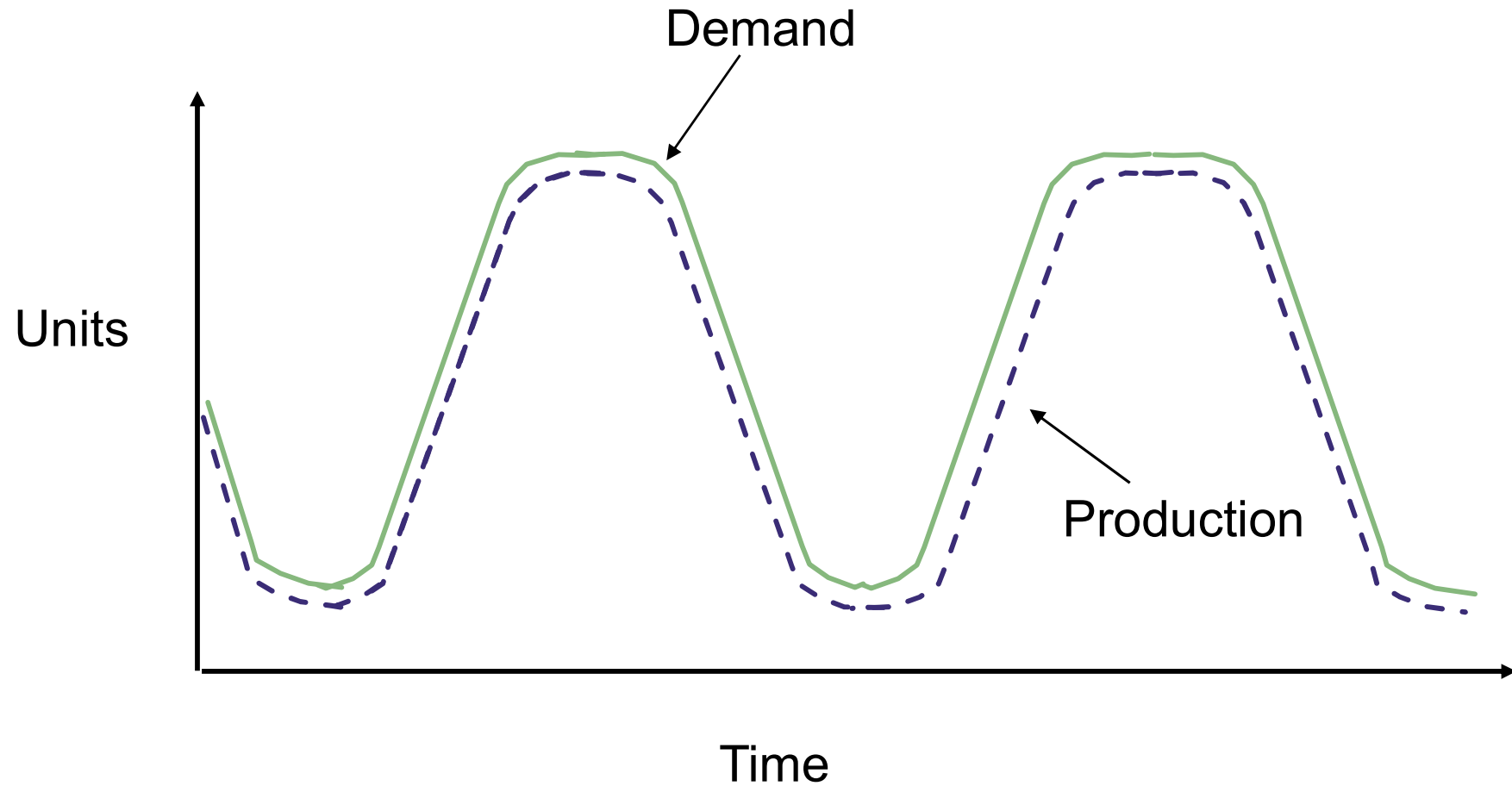
Level Production



Chase Demand

- Match output rates to demand forecast for each period by changing workforce levels or production rate
- **The disadvantages:** a ready supply of skilled labor may not always be available, newly hired personnel must be trained, and layoffs negatively affect the morale of all employees and may lead to a decrease in overall productivity.

Chase Demand



Mixed Strategies

- ◆ **Two or more options, such as overtime, subcontracting, hiring and layoff, etc., are used.**
- ◆ **There are both inventory changes and work force and production rate changes over the planning horizon.**
- ◆ **Typically, mixed strategies are better (result in lower costs) than pure strategies**

Demand and Capacity Options to Satisfy the Demand Over the Planning period

◆ **Proactive: Involve demand options:** Attempt to change demand to match capacity

- ✓ Influence the demand
- ✓ Backordering
- ✓ Producing counter-seasonal products

◆ **Reactive: Involve capacity options:** Attempt to change capacity to match demand

- ✓ Changing inventory Levels
- ✓ Varying workforce size by hiring or firing (layoffs)
- ✓ Varying production rate through overtime or idle time
- ✓ Subcontracting
- ✓ Using part-time workers

Capacity Options (Reactive)

1. Changing inventory levels

- ◆ **Increase** inventory in low demand periods to meet high demand in the future
 - ◆ **High inventory** may increase costs associated with storage, insurance, handling, obsolescence, and capital investment
 - ◆ **Low inventory** may cause shortages which may mean lost sales due to long lead times and poor customer service

Capacity Options(Reactive)

2. Varying workforce size by hiring or firing (layoffs)

- ◆ **Training and separation costs (benefit severance) for hiring and laying off workers**
- ◆ **New workers may have lower productivity**
- ◆ **Laying off workers may lower morale and productivity**

Capacity Options (Reactive)

3. Varying production rate through **overtime or idle time**

- ◆ Allows **constant workforce**
- ◆ May be difficult to meet large increases in demand
- ◆ **Overtime can be costly** and may drive down productivity
- ◆ Absorbing idle time may be difficult

Capacity Options

4. Subcontracting

- ◆ **Temporary measure during periods of peak demand**
- ◆ **May be costly**
- ◆ **Assuring quality and timely delivery may be difficult**
- ◆ **Exposes your customers to a possible competitor**

Capacity Options (Reactive)

5. Using part-time workers

- ◆ Useful for filling unskilled or low skilled positions, especially in services**

Demand Options (Proactive)

1. Influencing demand

- ◆ Use **advertising or promotion** to increase demand in low periods
- ◆ Attempt to shift demand to slow periods



Demand Options

2. Back ordering during high-demand periods

- ◆ **Requires customers to wait for an order without loss of goodwill or the order**
- ◆ **Most effective when there are few if any substitutes for the product or service**
- ◆ **Often results in lost sales**

Demand Options

3. Producing counterseasonal products

- ◆ **Develop a product mix of counter-seasonal items (furnaces and air conditioners)**
- ◆ **However, may lead to products or services outside the company's areas of expertise**

Roofing Supplier Example 1

(Graphical Approach)

Month	Expected Demand	Production Days	Demand Per Day (computed)
Jan	900	22	41
Feb	700	18	39
Mar	800	21	38
Apr	1,200	21	57
May	1,500	22	68
June	<u>1,100</u>	<u>20</u>	55
	6,200	124	

Table 13.2

$$\begin{aligned}
 \text{Average requirement} &= \frac{\text{Total expected demand}}{\text{Number of production days}} \\
 &= \frac{6,200}{124} = 50 \text{ units per day}
 \end{aligned}$$

Roofing Supplier Example 1

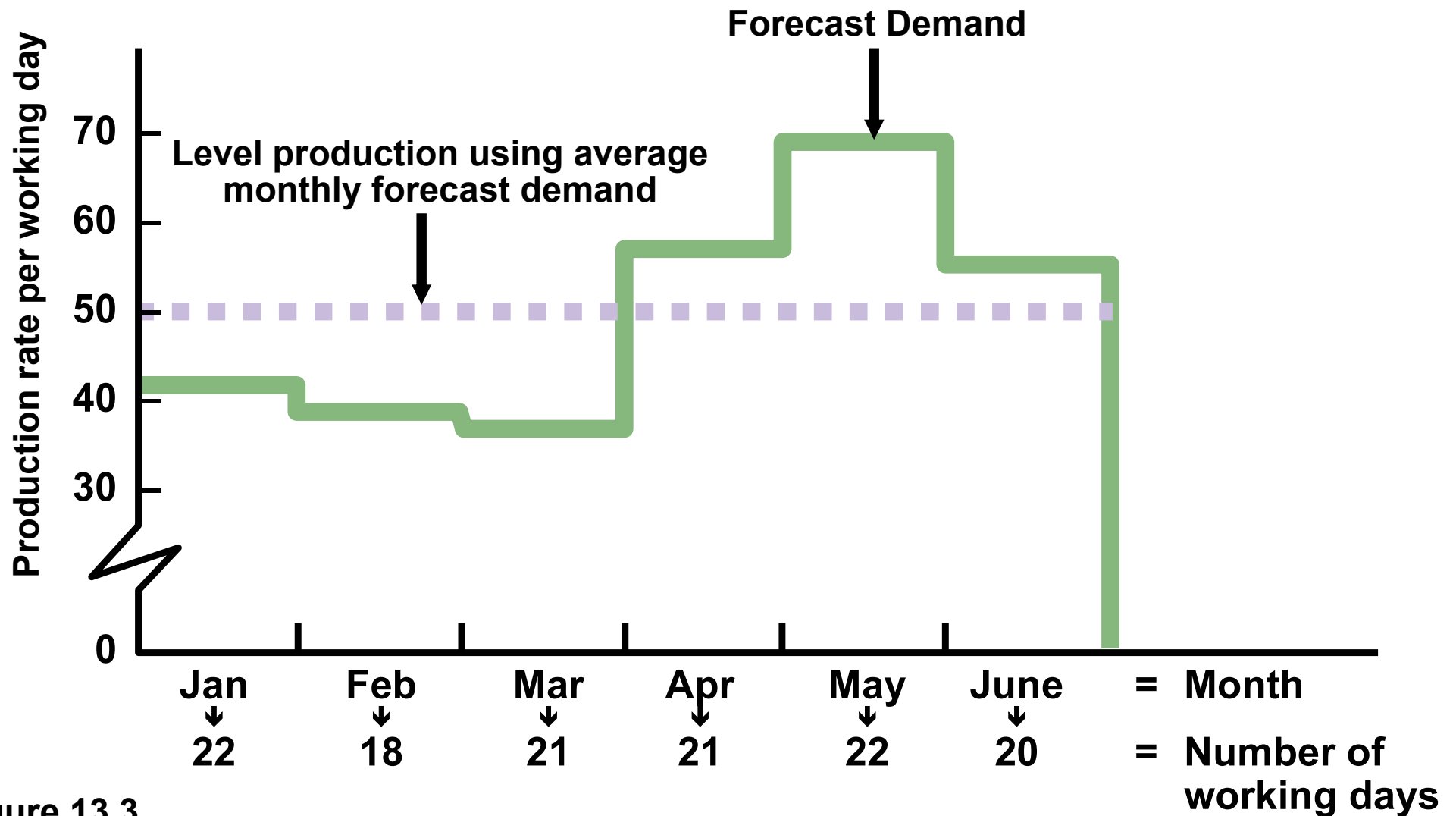


Figure 13.3

Roofing Supplier Example 1

Cost Information

Inventory carrying cost	\$ 5 per unit per month
Subcontracting cost per unit	\$20 per unit
Average pay rate (regular production)	\$10 per hour (\$80 per day)
Overtime pay rate	\$17 per hour (above 8 hours per day)
Labor-hours to produce a unit	1.6 hours per unit
Cost of increasing daily production rate (hiring and training)	\$300 per unit
Cost of decreasing daily production rate (layoffs)	\$600 per unit

Table 13.3

**Plan 1 – constant workforce
LEVEL PRODUCTION**

Roofing Supplier Example 1

Month	Production Days	Production at 50 Units per Day	Demand Forecast	Monthly Inventory Change	Ending Inventory
Jan	22	1,100	900	+200	200
Feb	18	900	700	+200	400
Mar	21	1,050	800	+250	650
Apr	21	1,050	1,200	-150	500
May	22	1,100	1,500	-400	100
June	20	1,000	1,100	-100	0
					1,850

Total units of inventory carried over from one month to the next = 1,850 units

Workforce required to produce 50 units per day = 10 workers

Workers Required = (50 unit/day*1.6 hr/unit)/8 hr/worker/day
=10 workers

Roofing Supplier Example 1

Costs		Calculations
Inventory carrying	\$9,250	(= 1,850 units carried x \$5 per unit)
Regular-time labor	99,200	(= 10 workers x \$80 per day x 124 days) or (6200 x 1.6 x 10)
Other costs (overtime, hiring, layoffs, subcontracting)	0	
Total cost	\$108,450	

Total units of inventory carried over from one month to the next = 1,850 units
 Workforce required to produce 50 units per day = 10 workers

Roofing Supplier Example 1

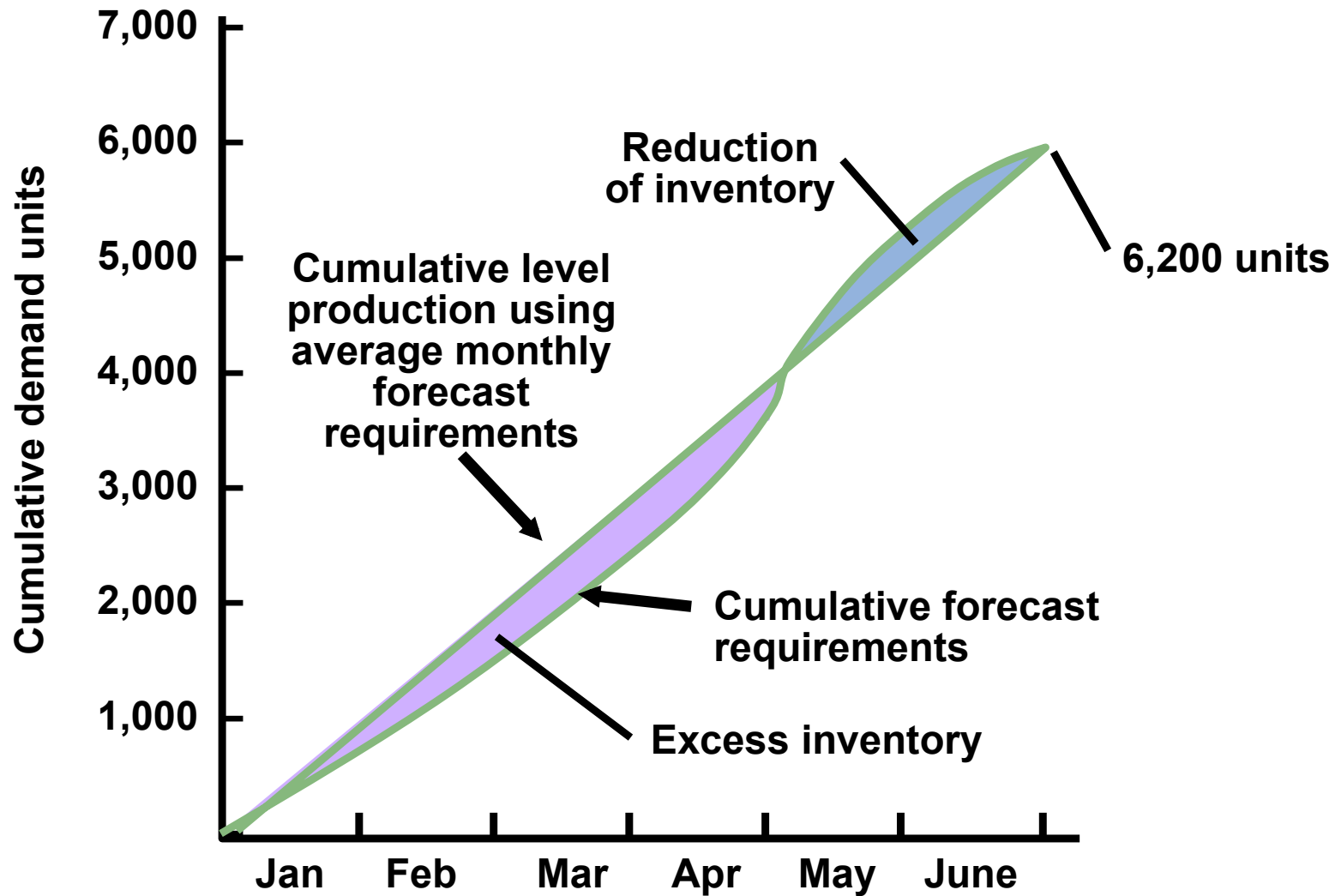


Figure 13.4

Roofing Supplier Example 2

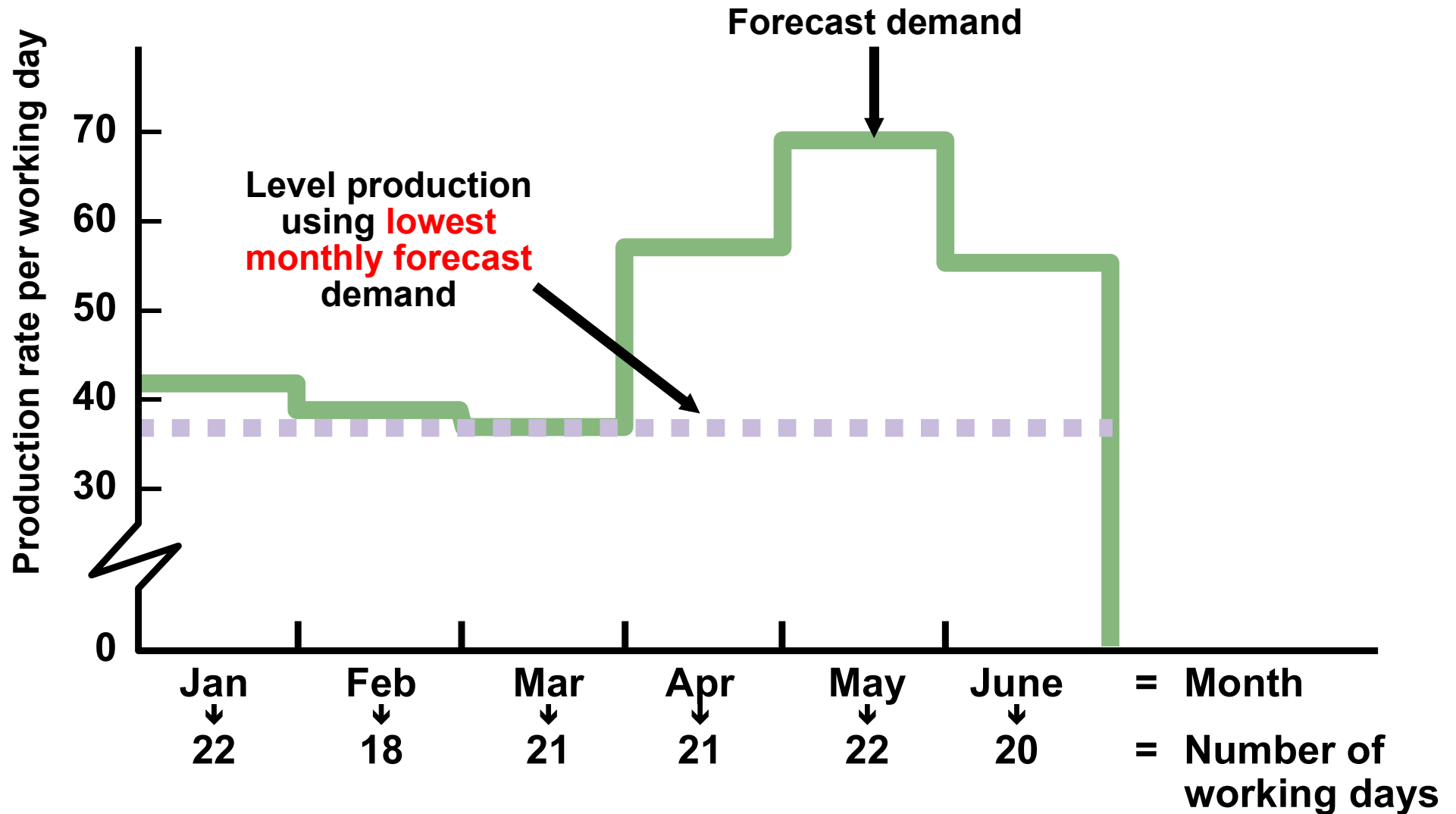
Month	Expected Demand	Production Days	Demand Per Day (computed)
Jan	900	22	41
Feb	700	18	39
Mar	800	21	38 MIN.
Apr	1,200	21	57
May	1,500	22	68
June	<u>1,100</u>	<u>20</u>	55
	6,200		

Table 13.2

2 – MIXED STRATEGY: Level the daily production minimum requirement over the planning horizon meet the excess demand with **subcontracting**

Minimum requirement = 38 units per day
 Constant work force = $(38 \times 1.6) / 8 = 7.6$ workers

Roofing Supplier Example 2



Roofing Supplier Example 3

Cost Information

Inventory carrying cost	\$ 5 per unit per month
Subcontracting cost per unit	\$20 per unit
Average pay rate	\$10 per hour (\$80 per day)
Overtime pay rate	\$17 per hour (above 8 hours per day)
Labor-hours to produce a unit	1.6 hours per unit
Cost of increasing daily production rate (hiring and training)	\$300 per unit
Cost of decreasing daily production rate (layoffs)	\$600 per unit

Table 13.3

Roofing Supplier Example 3

**In-house production = 38 units per day
x 124 days
= 4,712 units**

**Subcontract units = 6,200 - 4,712
= 1,488 units**

(layoffs)

Table 13.3

Roofing Supplier Example 3

**In-house production = 38 units per day
x 124 days
= 4,712 units**

Costs

Calculations

Regular-time labor

\$75,392

**(7.6 workers x \$80 per day x
124 days) or (4712 x 1.6 x
10)**

Subcontracting

29,760

(1,488 units x \$20 per unit)

Total cost

\$105,152

Roofing Supplier Example 3

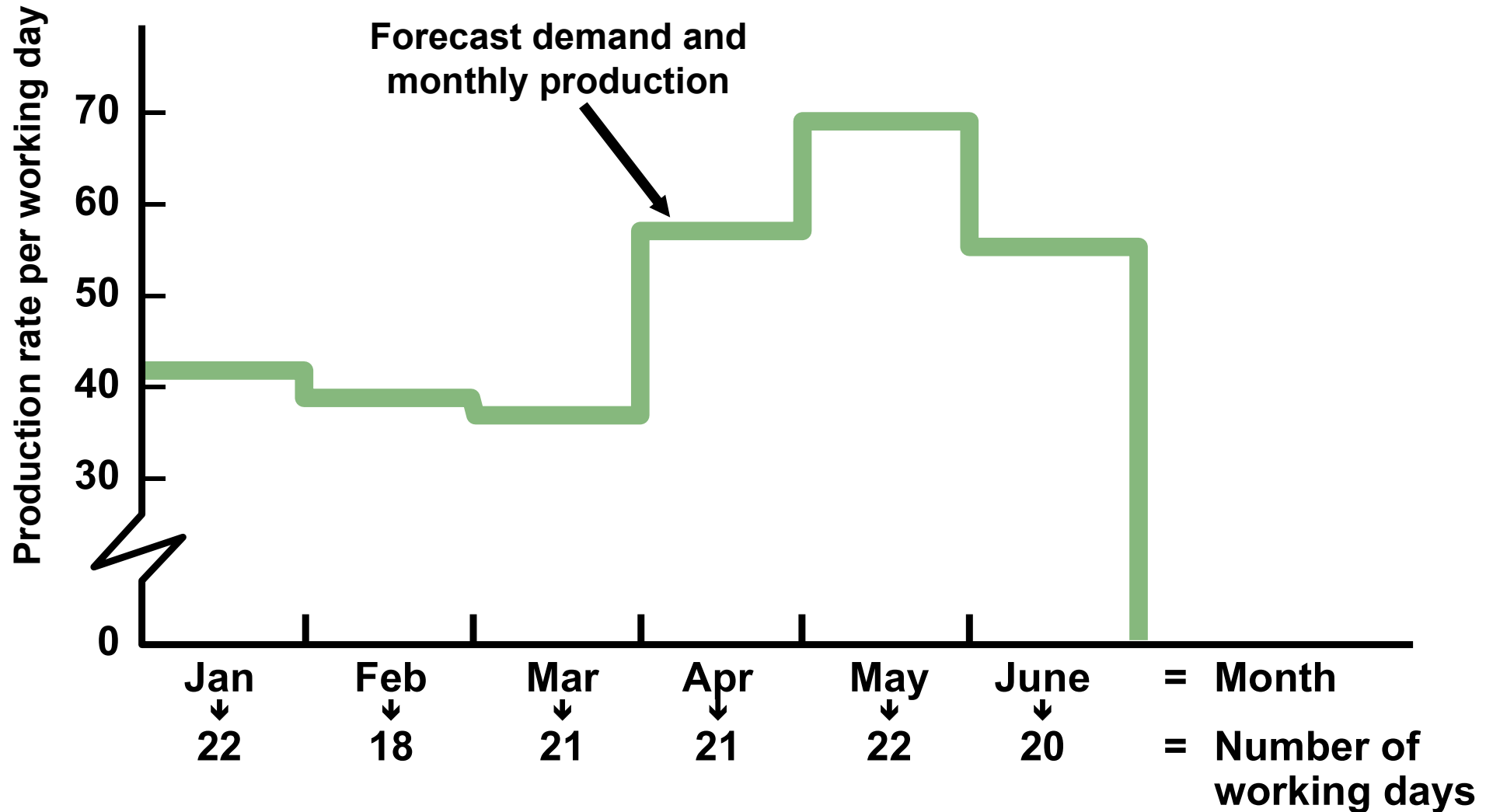
Month	Expected Demand	Production Days	Demand Per Day (computed)
Jan	900	22	41
Feb	700	18	39
Mar	800	21	38
Apr	1,200	21	57
May	1,500	22	68
June	<u>1,100</u>	<u>20</u>	55
	6,200	124	

Table 13.2

Plan 3 – hiring and layoffs
CHASE DEMAND

Production = Expected Demand

Roofing Supplier Example 4



Roofing Supplier Example 4

Cost Information

Inventory carrying cost	\$ 5 per unit per month
Subcontracting cost per unit	\$20 per unit
Average pay rate	\$10 per hour (\$80 per day)
Overtime pay rate	\$17 per hour (above 8 hours per day)
Labor-hours to produce a unit	1.6 hours per unit
Cost of increasing daily production rate (hiring and training)	\$300 per unit
Cost of decreasing daily production rate (layoffs)	\$600 per unit

Table 13.3

Roofing Supplier Example 4

Month	Forecast (units)	Daily Prod Rate	Regular Production Cost (demand x 1.6 hrs/unit x \$10/hr)	Extra Cost of Increasing Production (hiring cost)	Extra Cost of Decreasing Production (layoff cost)	Total Cost
Jan	900	41	\$ 14,400	—	—	\$ 14,400
Feb	700	39	11,200	—	\$1,200 (= 2 x \$600)	12,400
Mar	800	38	12,800	—	\$600 (= 1 x \$600)	13,400
Apr	1,200	57	19,200	\$5,700 (= 19 x \$300)	—	24,900
May	1,500	68	24,000	\$3,300 (= 11 x \$300)	—	24,300
June	1,100	55	17,600	—	\$7,800 (= 13 x \$600)	25,400
			<u>\$99,200</u>	<u>\$9,000</u>	<u>\$9,600</u>	<u>\$117,800</u>

Table 13.4

Comparison of Three Plans

Cost	Level	Mixed	Chase
Inventory carrying	\$ 9,250	\$ 0	\$ 0
Regular labor	99,200	75,392	99,200
Overtime labor	0	0	0
Hiring	0	0	9,000
Layoffs	0	0	9,600
Subcontracting	0	29,760	0
Total cost	\$108,450	\$105,152	\$117,800

Plan 2 is the lowest cost option

Table 13.5

Example 5

QUARTER	SALES FORECAST
Spring	80,000
Summer	50,000
Fall	120,000
Winter	150,000

Hiring cost = \$100 per worker

Firing cost = \$500 per worker

Inventory carrying cost = \$0.50 pound per quarter

Production per employee = 1,000 pounds per quarter

Beginning work force= 100 workers

a) Use Level Production Strategy

b) Use Chase Demand Strategy

Example 5

Level Production Strategy

QUARTER	SALES FORECAST
Spring	80,000
Summer	50,000
Fall	120,000
Winter	150,000

Level production

$$\frac{(50,000 + 120,000 + 150,000 + 80,000)}{4}$$

4

= 100,000 pounds

Example 5

Level Production Strategy

QUARTER	SALES FORECAST	PRODUCTION PLAN	INVENTORY
Spring	80,000	100,000	20,000
Summer	50,000	100,000	70,000
Fall	120,000	100,000	50,000
Winter	150,000	100,000	0
	Total	400,000	140,000

Cost = 140,000 pounds x 0.50 per pound = \$70,000

Example 5

Chase Demand Strategy

QUARTER	SALES FORECAST	PRODUCTION PLAN	WORKERS NEEDED	WORKERS HIRED	WORKERS FIRED
Spring	80,000	80,000	80	0	20
Summer	50,000	50,000	50	0	30
Fall	120,000	120,000	120	70	0
Winter	150,000	150,000	150	30	0
				100	50

Production per employee = 1,000 pounds per quarter

of required workers in spring = $80000/1000 = 80$ workers

Cost= (100 workers hired x \$100) + (50 workers fired x \$500)
= \$10,000 + 25,000 = \$35,000

Mathematical Approaches

- ◆ **Useful for generating strategies**
 - ◆ **Transportation Method of Linear Programming**
 - ◆ **Produces an optimal plan**
 - ◆ **Management Coefficients Model**
 - ◆ **Model built around manager's experience and performance**
 - ◆ **Other Models**
 - ◆ **Linear Decision Rule**
 - ◆ **Simulation**

Disaggregation of Aggregate Plan

- ◆ Working with aggregate units facilitates **intermediate planning**.
- ◆ But **to put this plan into action** we should decompose it, in other words, **disaggregate** it and state it in terms of actual units of products and plan the production for a shorter period of time
- ◆ Breaking down aggregate product to real specific products helps to calculate manpower, material and inventory requirements in detail.
- ◆ **Disaggregation results in a Master Production Schedule (MPS)** and MPS becomes input to Material Requirements Planning (**MRP**).

Master Production Schedule

- ◆ It indicates the quantity and timing of planned production by taking into account **desired delivery quantity** and timing as well as **on-hand inventory**.
- ◆ The **MPS** is one of the primary outputs of the **master scheduling process**.

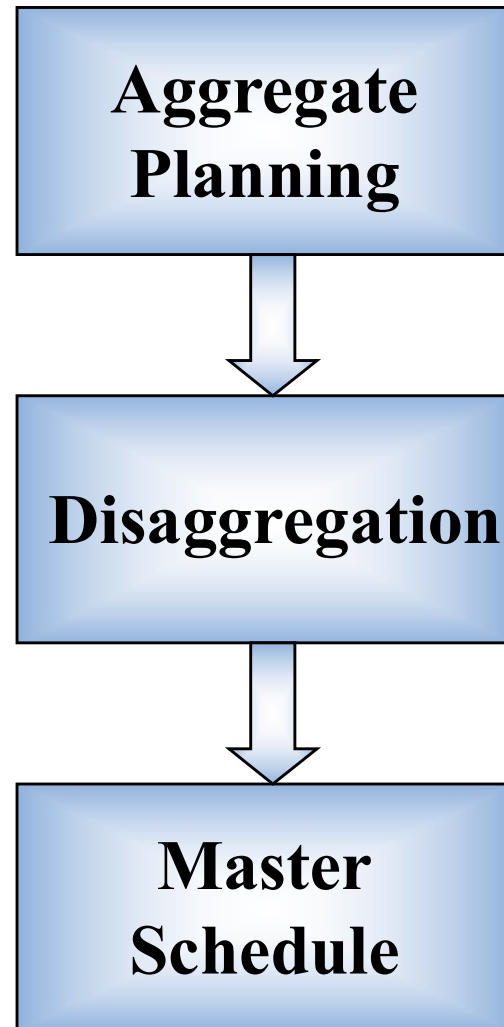
Rough-cut capacity Planning (RCCP)

- ◆ **RCCP involves testing the feasibility of a proposed MASTER SCHEDULE relative to available capacities, to assure that no obvious capacity constraints exist.**

Disaggregating the aggregate plan

- ◆ For example, televisions manufacturer may have an aggregate plan that calls for 200 television in January, 300 in February, and 400 in March.
- ◆ This company produces **21, 26, and 29 inch TVs**, therefore this three-month aggregate plan must be translated into specific numbers of TVs of each type (**Master Schedule**) prior to actually **purchasing the appropriate materials and parts, scheduling operations, and planning inventory requirements.**

Aggregate Plan to Master Schedule



Aggregate
plan

Jan	Feb	Mar.
200	300	400

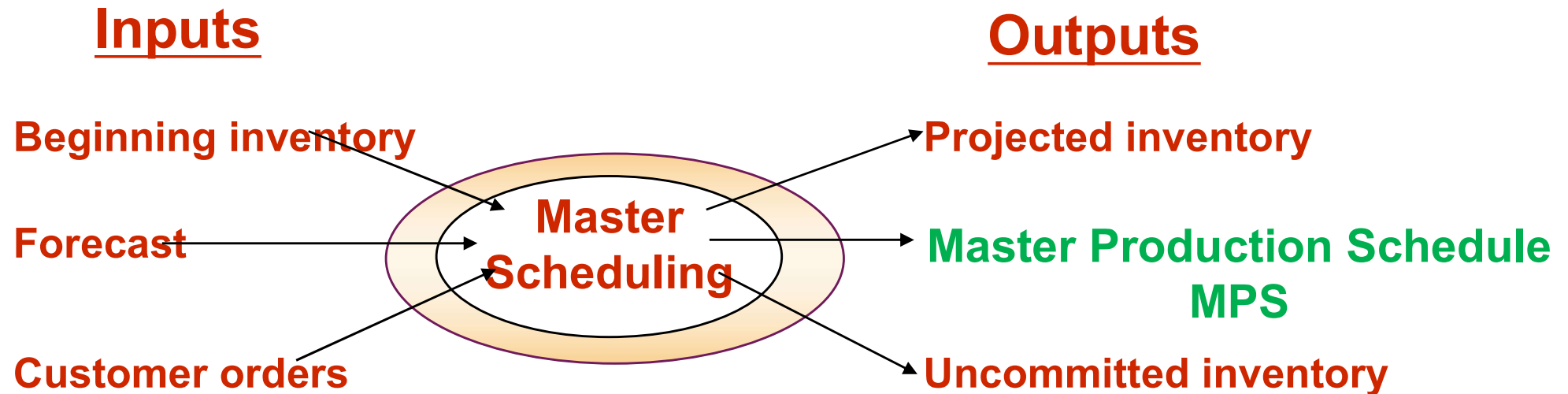
Master
schedule

Type	Jan.	Feb.	Mar
21 inch	100	100	100
26 inch	75	150	200
29 inch	25	50	100
total	200	300	400

Master scheduling

- ◆ The result of **disaggregating** the aggregate plan is a **master schedule** showing:
 - ✓ the quantity and timing of specific end items for a scheduled horizon, which often covers about **six to eight weeks** ahead.

Master Scheduling Process



Master schedule

◆ **Inputs:**

- ◆ **Beginning inventory; which is the actual inventory on hand from the preceding period of the schedule**
- ◆ **Forecasts for each period demand**
- ◆ **Customer orders; which are quantities already committed to customers.**

◆ **Outputs**

- ◆ **Projected inventory**
- ◆ **Production requirements (MPS)**

Example: Master Schedule

A company that makes industrial pumps wants to prepare a Master Production Schedule (MPS) for June and July.

- ✓ Marketing has forecasted demand of **120 pumps for June** and **160 pumps for July**.
- ✓ These have been evenly distributed over the four weeks in each month: **30 per week in June** and **40 per week in July**.

Example: Master Schedule

- ◆ Now suppose that there are currently **64 pumps in inventory (i.e., beginning inventory)**,
- ◆ There are customer orders that have been committed for the first five weeks (booked) and must be filled which are **33, 20, 10, 4, and 2** respectively.
- ◆ Suppose a **production lot size of 70** pumps is used.
- ◆ Prepare the Master Production Schedule

The master schedule before MPS

Beginning Inventory		JUNE				JULY			
	64	1	2	3	4	5	6	7	8
Forecast		30	30	30	30	40	40	40	40
Customer Orders (committed)		33	20	10	4	2			
Projected on-hand inventory		31	1	-29					

Forecast is larger than Customer orders in week 3

Customer orders are larger than forecast in week 1

Forecast is larger than Customer orders in week 2

Master Production Schedule (MPS)

Beginning inventory	June				July			
64	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Customer orders	33	20	10	4	2			
Inventory	31	1	41	11	41	1	31	61
MPS			70		70		70	70

Solution: The master schedule

- ◆ The first step you have to calculate the on hand inventory

Week	Inventory from previous week	Requirements	Net inventory before MPS	MPS	Projected inventory
1	64	33	31		31
2	31	30	1		1
3	1	30	-29	70	41
4	41	30	11		11
5	11	40	-29	70	41
6	41	40	1		1
7	1	40	-39	70	31
8	31	40	-9	70	61

Aggregate Planning Using Linear Programming

QUARTER	SALES FORECAST
Spring	80,000
Summer	50,000
Fall	120,000
Winter	150,000

Hiring cost = \$100 per worker

Firing cost = \$500 per worker

Inventory carrying cost = \$0.50 pound per quarter

Production per employee = 1,000 pounds per quarter

Beginning work force = 100 workers

APP by Linear Programming

$$\begin{aligned}\text{Minimize } Z = & \$100 (H_1 + H_2 + H_3 + H_4) \\ & + \$500 (F_1 + F_2 + F_3 + F_4) \\ & + \$0.50 (I_1 + I_2 + I_3 + I_4)\end{aligned}$$

Subject to

$$P_1 - I_1 = 80,000 \quad (1)$$

$$\text{Demand} \quad I_1 + P_2 - I_2 = 50,000 \quad (2)$$

$$\text{constraints} \quad I_2 + P_3 - I_3 = 120,000 \quad (3)$$

$$I_3 + P_4 - I_4 = 150,000 \quad (4)$$

$$\text{Production} \quad 1000 W_1 = P_1 \quad (5)$$

$$\text{constraints} \quad 1000 W_2 = P_2 \quad (6)$$

$$1000 W_3 = P_3 \quad (7)$$

$$1000 W_4 = P_4 \quad (8)$$

$$100 + H_1 - F_1 = W_1 \quad (9)$$

$$\text{Work force} \quad W_1 + H_2 - F_2 = W_2 \quad (10)$$

$$\text{constraints} \quad W_2 + H_3 - F_3 = W_3 \quad (11)$$

$$W_3 + H_4 - F_4 = W_4 \quad (12)$$

where

H_t = # hired for period t

F_t = # fired for period t

I_t = inventory at end
of period t

P_t = units produced
in period t

W_t = workforce size
for period t

Aggregate Planning for Services

- 1. Most services can't be inventoried**
- 2. Demand for services is difficult to predict**
(Some customers request prompt service or go elsewhere, if there is a waiting line)
- 3. Capacity is also difficult to predict**
- 4. Service capacity must be provided at the appropriate place and time**
- 5. Labor is usually the most constraining resource for services**

Yield Management

◆ **Yield management**

- ◆ **An approach to maximizing revenue by using a strategy of variable pricing; prices are set relative to capacity availability**
- ◆ **During periods of low demand, price discounts are offered**
- ◆ **During periods of peak demand, higher prices are charged**
- ◆ **Users of yield management include**
 - ◆ **Airlines, restaurants, hotels, restaurants**

••• **13.11** Liz Perry Health and Beauty Products has developed a new shampoo and you need to develop its aggregate schedule. The cost accounting department has supplied you the cost relevant to the aggregate plan and the marketing department has provided a four-quarter forecast. All are shown as follows:

Quarter	Forecast
1	1,400
2	1,200
3	1,500
4	1,300

	Costs
Previous quarter's output	1,500 units
Beginning inventory	0 units
Stockout cost for backorders	\$50 per unit
Inventory holding cost	\$10 per unit for every unit held at the end of the quarter
Hiring workers	\$40 per unit
Layoff workers	\$80 per unit
Unit cost	\$30 per unit
Overtime	\$15 extra per unit
Subcontracting	Not available

Your job is to develop an aggregate plan for the next four quarters.

- First, try a chase plan by hiring and layoffs (to meet the forecast) as necessary.
- Then try a plan that holds employment steady.
- Which is the more economical plan for Liz Perry Health and Beauty Products? 

••• **13.12** Missouri's Soda Pop, Inc., has a new fruit drink for which it has high hopes. Steve Allen, the production planner, has assembled the following cost data and demand forecast:

Quarter	Forecast
1	1,800
2	1,100
3	1,600
4	900

n

Costs/Other Data

Previous quarter's output = 1,300 cases

Beginning inventory = 0 cases

Stockout cost = \$150 per case

Inventory holding cost = \$40 per case at end of quarter

Hiring employees = \$40 per case

Terminating employees = \$80 per case

Subcontracting cost = \$60 per case

Unit cost on regular time = \$30 per case

Overtime cost = \$15 extra per case

Capacity on regular time = 1,800 cases per quarter

Steve's job is to develop an aggregate plan. The three initial options he wants to evaluate are:

- Plan A: a chase strategy that hires and fires personnel as necessary to meet the forecast.
- Plan B: a level strategy.
- Plan C: a level strategy that produces 1,200 cases per quarter and meets the forecasted demand with inventory and subcontracting.

- a) Which strategy is the lowest-cost plan?
- b) If you are Steve's boss, the VP for operations, which plan do you implement and why? 